

A. GENERAL NOTES

1. All construction shall conform to the Design Codes in Section "C. Design Criteria," including all applicable standards and documents referenced within those codes.
2. Plan and detail notes provided on specific sheets within these drawings supplement information in these General Notes. Always coordinate the requirements of these notes with what is shown within the drawings.
3. Unless noted specifically on a plan, all floor plans show framing for the floor indicated and vertical framing (walls, openings, posts, columns) above that floor. The roof plan shows roof framing only.
4. Contract Document Coordination:
- a. The drawings contained herein are intended to be utilized in conjunction with other design consultant's drawings (architectural, civil, mechanical, etc.). It is the responsibility of the Contractor to coordinate the requirements of the drawings into their shop drawings and construction.
 - i. Refer to the Project Specifications issued as part of the contract documents for information supplemental to these drawings.
 - Should conflicts between these drawings and the Specifications exist, the Contractor shall bring them to the attention of the structural engineer for clarification.
 - b. Refer to the architectural, mechanical, electrical, and civil drawings for location and size of block outs, inserts, openings, curbs, bases & pads, and dimensions not shown on these drawings.
 - c. Refer to the architectural drawings for size and location of doors and window openings, exterior wall assemblies, and floor, wall, and roof finishes. Refer to the mechanical and electrical drawings for additional information including locations of mechanical units, generators, etc.
 - d. Omissions or conflicts between various elements of the drawings, notes and details shall be brought to the attention of the structural engineer and resolved before proceeding with the work.
5. Use of Drawings in Construction:
- a. The Contractor shall verify all dimensions and conditions at the job site before commencing work and shall report any discrepancies to the engineer responsible for the design of that work.
 - b. Do not use scaled dimensions; use written dimensions or, where no dimension is provided, consult the structural engineer for clarification before proceeding with the work.
 - c. Details and keynotes shown shall be incorporated into the project at all appropriate locations, whether or not they are specifically referenced on the drawings.
 - d. McClure may provide the contractor with electronic files for their convenience and use in the preparation of shop drawings. These electronic files are not construction documents; the contractor is not relieved of his/her duty to fully comply with the contract documents, including the need to confirm and coordinate all dimensions and details, take field measurements, verify field conditions, and coordinate the contractor's work with that of other contractors for the project.
6. Changes During Construction:
- a. Openings shall not be cut or otherwise made in any structural member unless that opening is specifically shown on these drawings. The Contractor shall seek approval in writing from the structural engineer for any design incorporating additional openings.
 - b. Support details shown for Architectural, Mechanical, Electrical, and Plumbing equipment as well as elevators is based upon available information from the manufacturer (if any). The Contractor shall coordinate requirements of actual equipment supplied with details and shall provide any additional framing required.
 - c. The Contractor has the responsibility to notify the structural engineer of any architectural, mechanical, electrical, or plumbing load imposed on the structure that is not documented on the Contract Documents or differs from what is originally shown. Provide documentation of location, load, size, and anchorage of all undocumented loads in excess of 250 lbs.
7. Construction Sequence and Methods
- a. These drawings and the related Specifications represent the finished structure and, except where specifically shown, do not indicate the method or means of construction. Loads on the structure during construction shall not exceed the design loads indicated in Section "A. Design Criteria" as a maximum. The Contractor shall supervise and direct the work and shall be solely responsible for all construction means, methods, procedures, techniques, and sequence.
 - b. The Contractor is responsible for compliance with all applicable job-related safety standards proceeding from governing organizations (e.g. OSHA).
 - c. It is the responsibility of the Contractor to ensure the stability of the structural elements during construction as a result of means and sequence by providing shoring, bracing, etc. as required.
 - i. Stability considerations should include all applicable temporary construction and environmental loads per ASCE 37 which may include wind and seismic forces.
 - ii. Temporary bracing shall remain in place until positive connection is made between the braced element and the floor/roof diaphragm or foundation above and below, and those diaphragms in turn are structurally connected and connected to the vertical elements of the lateral force resisting system. This is a means and methods item.
 - iii. The Contractor may at their discretion employ a Specialty Structural Engineer, licensed in the state where the project is located, for the design of any temporary bracing, lifting, rigging, and shoring. Any sealed drawings, calculations, reports, etc. prepared for construction stability shall be submitted to the structural engineer for review.
 - d. The Contractor shall consider the effects of thermal movements due to hot or cold weather construction and the potential for extreme temperature variations before the structure is complete.
 - e. Any foundation wall restrained by a floor is not designed to be backfilled prior to the complete construction of the floor and the lateral bracing elements (shear walls, braced frames, etc.) below it. For backfilling before this time, temporary bracing shall be designed and provided by the Contractor.
 - f. Where specifically indicated on the drawings, walls may be backfilled according to instructions provided before the final connection to the slab is made.
 - g. The Contractor is responsible for the protection and repair of any adjacent existing structures, surfaces, and areas which may be damaged as a result of the work.

B. STRUCTURAL ENGINEERING DESIGN NARRATIVE

1. McClure Engineering Company (McClure, MEC) is the Structural Engineer of Record (EOR) responsible for the documentation of structural design criteria, strength and stability of the primary vertical and lateral load-carrying systems in their completed form, and conformance of the structural design to the applicable building codes. These drawings produced by McClure convey the structural engineering design for the project, which includes the following components and systems:
- a. Foundations consisting of trench footings and isolated column footings.
 - b. Slabs on grade.
 - c. Residential tower framing above the slab on grade consisting of:
 - i. Load-bearing wood wall and opening framing.
 - ii. Gypcrete over wood T&G Sheathing over wood joists, floor and roof trusses.
 - d. Structural steel framing identified on the drawings.
 - e. The lateral force resisting system of the structure consisting of sheathed gypsum and wood shear walls and wood sheathed diaphragms.
2. The following items are Deferred Submittals. Framing intent and additional requirements for these structural components are provided within these drawings":
- a. Structural steel connections – see general notes section "Structural Steel"
 - b. Wood Floor & Roof Trusses – see general notes section "Wood Framing and Fastening" | see S001 for applicable design criteria.
 - c. Connections of Wood Floor & Roof Trusses to the supporting structure."
 - d. All premanufactured canopy and awning framing including connections to the structure.
- * Reference section "D. Submittal Requirements." Coordinate requirements of these drawings with those of other design consultant drawings and the Project Specifications.
3. The following items are specifically excluded from McClure's design scope as represented on these drawings:
- a. Requirements for fire rating of assemblies or fire protection of structural members.
 - b. Global stability of soil mass.
 - c. Any exterior slabs, bollards, curbs, and any enclosures not shown on these drawings.
 - d. Interior non-load-bearing wood wall or ceiling framing.
 - e. Shoring design, formwork design, temporary bracing, and other means and methods items.

C. DESIGN CRITERIA

1. Design Codes:
- a. International Building Code: IBC 2018
 - b. Minimum Design Loads for Buildings and Other Structures: ASCE 7-16
2. Design Loads:
- a. Dead Loads
 - Floors = 35 psf
 - Floors (Corridor) = 30 psf
 - Exterior Walls (Stone Siding) = 47 psf
 - Exterior Walls (Plank Siding) = 13 psf
 - Roof = 22 psf plus mechanical equipment shown on roof plan
 - b. Live Loads (reducible per code UNO)
 - Slab on Grade = 125 psf (non-reducible)
 - Residential = 40 psf
 - Corridors/Public Areas = 100 psf
 - Mechanical/Storage = 125 psf (non-reducible)
 - Typical Roof = 20 psf
 - c. Roof Snow Load
 - Ground Snow Load (p_g) = 30 psf (per City of Carter Lake, IA)
 - Flat Roof Snow Load (p_f) = 21 psf
 - Snow Exposure Factor (C_e) = 1.0
 - Snow Load Importance (I_s) = 1.0
 - Thermal Factor (C_t) = 1.0
 - Slope Factor (C_s) = 1.0
 - Snow Drift (Parapets) = 6 psf
 - Surcharge = 2.75 ft
 - Drift Width = 47 psf
 - Snow Drift (Low Roof) = 10.5ft (Leeward Snow Drift from Higher Roof)
 - Surcharge = 47 psf
 - Drift Width = 10.5ft (Leeward Snow Drift from Higher Roof)
 - d. Wind Load
 - Basic Design Wind Speed, V = 105 mph (3 sec. Gust) (per City of Carter Lake, IA)
 - ASD Wind Speed, V_{asd} = 81.3 mph
 - Risk Category = II
 - Wind Exposure = C (per City of Carter Lake, IA)
 - Internal pressure Coefficient (GC_p) = ±0.18
 - Components and Cladding (psf):

Zone	A=10ft ²	A=50 ft ²	A=100 ft ²
1	+16/-46	+16/-39	+16/-36
2	+16/-60	+16/-51	+16/-47
3	+16/-82	+16/-64	+16/-56
4	+26/-29	+24/-26	+23/-25
5	+26/-35	+24/-30	+23/-27
 - Notes:
 - 1. A is the Effective Wind Area as defined in ASCE 7 Ch. 28.
 - 2. Linear interpolation between tabulated values is permitted.
 - 3. Elements with Tributary Area (A_t) > 700 ft² shall be permitted to be designed using provisions for MWFRS.
 - e. Earthquake Load
 - Risk Category = II
 - Seismic Importance Factor (I_s) = 1.0
 - S_s = 0.071g S₁ = 0.045g
 - Soil Site Class: D (Code Default)
 - S_{DS} = 0.076 S_{DI} = 0.072
 - Seismic Design Category = B (per City of Carter Lake, IA)
 - Basic Seismic Force Resisting System(s):
 - Light-Frame Walls with shear panels – all other materials (ASCE 7 Table 12.2-1 Line A.17)
 - R = 2.0 Ω₀ = 2.0 C_u = 2.0 C_s = 0.038
 - Design Base Shear, V = C_s x W
 - Analysis Procedure = Equivalent Lateral Force Procedure (ASCE 7-16 Chapter 12.8)
 - f. Rain Load
 - Rain Intensity (i) = 7.07 in/hr

	Total Load	Live/Snow/Wind Load	Absolute Maximum
Floor Joists/Trusses	L360	L480	1"
Roof Joists/Trusses	L240	L260	1.5"
Wall Framing (flexible finish)		L240	0.75"
Wall Framing (brittle/brick finish)		L360	0.5"

Cantilever deflection limits are the more restrictive of 2 x the appropriate L/-- limit (e.g. 2L/360 = L/180) or absolute maximum value listed above, measured at the tip of the cantilever U.N.O.

4. Soil Properties:
- a. Soil properties are based on the project geotechnical report titled Geotechnical Engineering Report, prepared by ISG-TEAM, Inc. on August 1, 2025 (herein known as "Geotechnical Report").
 - b. Allowable Soil Bearing Pressure = 2000 psf
Note that overexcavation may be required to achieve this allowable bearing pressure per Geotech Report section "Shallow Foundation Construction" for all foundations. Verify with Geotechnical Representative.
 - c. Minimum Embedment below Finished Grade = 42 inches

D. SUBMITTAL REQUIREMENTS

1. Submittal Procedures:
- a. The Contractor shall provide all submittals in PDF format unless otherwise requested or indicated in the Project Specifications.
 - b. All submittals must be reviewed by the Contractor prior to McClure's review. The Contractor is responsible for reviewing each submittal for basic coordination with these drawings and to verify that all the required components of the submittal are incorporated. The submittal must bear the electronic review stamp of the Contractor before McClure will proceed with the review.
 - c. Incomplete submittals or submittals not meeting the requirements of this section will not be reviewed. McClure will notify the contractor that the submittal is incomplete or unacceptable and that resubmission is required.
 - i. Submittals requiring engineering calculations for all or a portion of the work are considered incomplete without the sealed calculations and will not be reviewed.
 - ii. Shop Drawings shall be original drawings. Submissions incorporating any portion or reproduction of the contract documents will not be reviewed.
 - iii. Deferred Submittals not meeting the seal requirements of section D.2.b are considered incomplete and will not be reviewed.
 - iv. Resubmittals with comments from a previous review left unaddressed or without any response will not be reviewed.
 - d. Allow two weeks for review of all submittals unless an agreement for expedited review is made in writing by McClure.
 - e. McClure's submittal review scope of work includes a single submittal review and one review of the revised submittal if required (two reviews total of the same submittal). Time required for more than two reviews of a submittal is considered an additional service and will be billed hourly. McClure reserves the right to withhold review of a submittal surpassing this allowance until proper billing to the responsible party can be established.
 - f. Submittals must be returned to the Contractor by McCure bearing a stamp marked "Reviewed No Exception Taken" or "Reviewed With Comments/Exceptions" prior to proceeding with the work. Submittals marked "Reject/Resubmit" must be revised according to the comments provided prior to commencing with the respective scope of work.
2. Deferred Submittals:
- a. See Section "B. Structural Engineering Design Narrative" for the list of items considered Deferred Submittals.
 - b. Deferred Submittals shall bear the seal of a professional engineer licensed in the state where the project is located. If the project requires a licensed Structural Engineer (S.E.) as the Engineer of Record according to state laws, the same qualification level applies to the engineer sealing the Deferred Submittals.
 - c. Deferred Submittal items shall not be installed until the Deferred Submittal documents have been approved by the Building Official.
3. Submittal List:
- a. Submittals (product data, test records, shop drawings, and/or calculations) are required for the following:

Submittal Name	Items Required:			
	Product Data	Shop Drawings	Test Records	Engineering Drawings
1. Concrete Mix Designs	X		X	
2. Concrete Break Reports			X	
3. Concrete Reinforcing Layout		X		
4. Concrete Anchor Bolts & Embedded Plates	X	X		
5. Concrete & CMU Anchors (Post-Installed)	X			
6. Post-Installed Anchor Substitutions				X
7. Post-Installed Connection Geometry Alteration	X		X	X
8. Masonry Wall Materials	X		X	
9. Masonry Reinforcing		X		
10. Brick & Stone Veneer	X			
11. Structural Steel Framing	X	X		
12. Structural Steel Framing Connections		X		X
13. Metal Canopies & Awnings	X	X		X
14. Wood Framing Materials	X			
15. Wood Floor & Roof Trusses incl. Reactions			X	
16. Wood Truss Connections to Supporting Structure			X	X
17. Specialty Wood Fasteners	X			

- b. "Product Data" may indicate mill certifications, material data sheets, Evaluation Service Reports (ESRs), etc. See requirements of each

E. CONCRETE

1. Reinforced concrete shall have the following minimum 28 day compressive strengths:
- a. Slab on grade, unless noted otherwise 4000 psi normal weight
 - b. Foundations 5000 psi normal weight
2. All concrete exposed to weather shall have 6% (+/- 1%) air entrainment.
3. Submit mix designs for all concrete mixes prior to placement. All submittals shall include the following:
- a. Batch quantities including admixture dosage rates.
 - b. Strength test results for trial mixes.
 - c. Aggregate source(s) and gradation(s).
 - d. Product data for cement, fly ash and other cementitious materials.
 - e. Product data for all admixtures.
4. Provide protection for reinforcing bars as follows:
- a. Concrete cast against and permanently exposed to earth 3"
 - b. Concrete exposed to earth and weather (formed)
 - i. #5 and smaller 1-1/2"
 - ii. #6 and larger 2"
 - c. Concrete not exposed to weather and not in contact with ground:
 - i. Slabs and walls 3/4"
 - ii. Beams and columns 1-1/2"
5. Interface of all slab and beam construction joints shall be roughened with 1/4" amplitude. Surface of construction joints shall be clean and free of laitance. Immediately before new concrete is placed, construction joints shall be wetted and standing water removed.
6. Construction joints in walls shall be keyed and placed at locations approved by the Architect and Structural Engineer.
7. Provide control joints in all retaining walls at 15 ft to 20 ft intervals.
8. Provide PVC waterstops in all below grade construction joints and at other locations as shown.
9. Provide compressible filler and sealant in all slab-on-grade and wall and column interfaces that are not dowelled together.
10. All column pockets shall be filled with concrete after column is erected.
11. Sleeves and openings in slabs not shown on structural drawings or outside the parameters of typical sleeve details are not permitted, unless approved by the Structural Engineer.
12. Conduit and pipes embedded in slabs, walls, or grade beams shall be no larger in outside dimension than 1/3 the overall member thickness and shall be placed no closer than 3 diameters or widths on center.
13. Provide concrete housekeeping pads under all mechanical, plumbing, fire protection, and electrical equipment per plans. Pads shall extend beyond equipment a nominal 6" on all sides. Apply a bonding agent to existing concrete slab prior to pouring of housekeeping pad. Provide reinforcing per details.
14. At floor drains, locally slope floor towards drain. See architectural and plumbing drawings for drain locations.
15. Foundation walls shall be temporarily braced until positive attachment is made to floor framing per details. This is a means and methods item.

Slab on Grade

- 1. Slab shall be constructed as shown on plans.
- 2. Slab-on-grade shall be founded on 8" deep 3/4" clean aggregate base.
- 3. The upper 18" of subgrade shall consist of low volume change material such as rollstone or wastelime. Granular fill shall be compacted to a minimum of 95% of the ASTM D698 maximum dry Standard Proctor density. The 6" aggregate base shall be included in the 18" depth required for the low volume change layer.
- 4. Provide joints at 30 x slab thickness (+/-) in both directions and located to conform to bay spacing wherever possible (at column centerlines, half bays, third bays, etc.). Submit control joint layout to Architect for any exposed concrete surface.
- 5. Saw cut control joints shall be done late enough to prevent raveling of the cut edges and early enough to prevent cracking of the slab ahead of the saw blade.
- 6. Concrete slab to be cured according to ACI Standards. Concrete slab cure to be compatible with any sealer, grout, or adhesive that may be used on the floor later.
- 7. At floor drains, locally slope floor towards drain. See architectural and plumbing drawings for drain locations.

Subsurface Requirements

- 1. Foundation design is based on geotechnical report by ISG-TEAM, Inc, dated 08/01/2025.
- 2. A geotechnical representative shall be retained on site for all construction activity to verify that all proper requirements have been met to meet the design requirements outlined in the geotechnical report. Representative shall be ISG-TEAM, Inc. or someone familiar with all documents of the geotechnical investigation provided for the project.
- 3. The Contractor shall provide dewatering of excavations from surface water and ground water. Do not place concrete if water is present at base of excavation.
- 4. Geotechnical Testing Agency Requirements
 - a. If the geotechnical representative on site takes exception to anything in the Geotechnical Report and requires additional field investigation to clarify those expectations, the cost of such investigation shall be included in the additional fee for field quality control and testing and identified as such. All other exceptions, the cost of such investigation shall be included in the additional fee for field quality control and testing and identified as such. All other exceptions shall be documented and approved by the geotechnical engineer.
 - b. The geotechnical representative must have read all documents pertaining to the geotechnical report for the project and understood and accepted the criteria contained in the report.
- 5. The geotechnical representative must understand and be able to make decisions affecting the work for field observations and conditions described in the report during construction. The representative must be capable of advising the owner or contractor for procedures regarding, but not limited to: sub-grade preparation, dewatering activities, and other construction considerations.

F. REINFORCING FOR CONCRETE

1. General
- a. All reinforcing steel to be ASTM A615, Grade 60, deformed bars, unless noted otherwise.
 - i. Any reinforcing to be welded shall be ASTM A706 and welded with E80 electrodes.
 - ii. Alternatively, ASTM A615 reinforcing may be welded with E80 electrodes and proper preheat according to AWS D14.1.
 - iii. E70 electrodes are not permitted for welding rebar.
 - b. Welded wire fabric shall be plain wire conforming to ASTM A1064. Welded wire fabric shall be in flat sheets.
 - c. All reinforcing bars to be detailed and placed in accordance with the ACI Manual of Standard Practice for Detailing Reinforced Concrete Structures' specifications
 - d. All reinforcing, including dowels, shall be securely tied and cast with the lower member. Placing reinforcing after concrete has been placed will not be permitted.
 - e. Field bending of reinforcing partially embedded in concrete will not be allowed unless specifically noted on the drawings or approved by the Structural Engineer.
 - f. All reinforcing bars shall be contact lap spliced or doweled as follows, unless noted otherwise:

Tension Development and Splice Lengths for f _y = 4,000psi									
Bar Size	Development			Class "B" Splice		Standard 90 deg. Hook			
	Top Bar	Other Bar	Other Bar	Top Bar	Other Bar	Embed	Leg	Bend	Dia.

Bar Size	Development			Class "B" Splice		Standard 90 deg. Hook			
	Top Bar	Other Bar	Other Bar	Top Bar	Other Bar	Embed	Leg	Bend	Dia.
#3	17	13	22	17	6	8	6	2-1/4	
#4	22	17	29	22	6	8	8	3	
#5	28	22	36	28	8	10	10	3-3/4	
#6	33	26	43	33	9	12	14	4-1/2	
#7	49	37	63	49	11	14	17	5-1/4	
#8	55	43	72	55	12	16	16	6	
#9	63	48	81	63	14	19	19	9-1/2	

- 1. Straight development and Class "B" splice lengths shown in above tables are based on uncoated bars assuming center-to-center bar spacing ≥ 3'd, without ties or stirrups or ≥ 2'd, with ties or stirrups, and bar clear cover ≥ 1.0'd, Normal weight concrete as well as no transverse reinforcing are both assumed.
- 2. Standard 90 deg. hook embedment lengths are based on bar side cover ≥ 2.5" and bar end cover ≥ 2" without ties around hook.
- 3. For special seismic considerations, refer to ACI 318 Code Chapter 21.
- 4. All tension splices shall be Class "B" splices unless noted otherwise on plans.

- g. All welded wire fabric shall be lapped 12" or 48 wire diameters, whichever is greater.
- h. Provide (2) #5 x 6'-0" diagonals at all corners of openings and re-entrant corners, unless noted otherwise.
- i. Dowels between foundation and walls shall be installed and shall be the same grade, size, and spacing as the vertical wall reinforcing unless noted otherwise.
- j. Provide corner bars to match longitudinal reinforcing in all footings. Provide (2) corner bars at tee intersections.
- k. Provide 250 pounds of miscellaneous straight bar reinforcing (#4 & #5) to be used in field for special conditions. Labor for placing same to be included.
- 2. Slabs and Slabs-on-Grade
 - a. All slabs on grade to be reinforced with 6x6 – W2.9xW2.9 welded wire fabric, unless noted otherwise.
- 3. Walls
 - a. Provide corner bars in the outside face and at wall intersections to match horizontal wall bars. Use (3) #5 vertical construction rods a corners.
 - b. Minimum reinforcing shall be as follows for each wall thickness, unless noted otherwise:
 - 6" wall – #4@16 one layer
 - 8" wall – #4@12 one layer



NOTICE:
McClure Engineering Co. is not responsible or liable for any issues, claims, damages, or losses (collectively, "Losses") which arise from failure to follow these Plans, Specifications, and the engineering intent they convey, or for Losses which arise from failure to obtain and/or follow the engineers' or surveyors' guidance with respect to any alleged errors, omissions, inconsistencies, ambiguities, or conflicts contained within the Plans or Specifications.

IOWA CERTIFICATE OF AUTHORITY
NO. 26887



MARCUS HIMMELBERG
NO. 24622
EXP. 12/31/2025

I HEREBY CERTIFY THAT THIS ENGINEERING DOCUMENT WAS PREPARED BY ME OR UNDER MY DIRECT PERSONAL SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF IOWA.

No.	Description	Date

PROJECT NUMBER: 2025001420
SET ISSUE DATE: 08/01/2025
ENGINEER: IWC
DRAWN BY: CEL
CHECKED BY: MDH

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